

WTW 158 Worksheet 4.1 - 4.3 2016

1. Find the extreme values of the following functions / Vind die ekstreme waardes van die volgende funksies:

(a) $f(x) = \frac{2x^2}{(x^2-1)}$ over the interval $[-\frac{1}{2}, \frac{3}{4}]$

(b) $f(x) = \frac{2x^2}{(x^2-1)}$ over the interval $[-\frac{1}{2}, 2]$

(c) $f(x) = (x+1)^5 - 5x - 2$ over the interval $[-2, 2]$

2. Determine the intervals where each of the following functions are increasing or decreasing if $f(x)$ is given by: / Bepaal die intervale waarop elkeen van die volgende funksies stygend of dalend is as $f(x)$ gegee word deur:

(a) $x^2 - 4x - 6$ (b) $x + \frac{1}{x}$ (c) x^3 .

3. Use the 1st and 2nd derivative test to determine the local minimum as well as the local maximum of the following functions. Vind die lokale minimum en maksimumwaardes van die onderstaande funksies $f(x)$ mbv die 1e sowel as die 2e afgeleide toets.

(a) $x^2 - 4x - 6$ (b) $x + \frac{1}{x}$ (c) x^3 (as above).

4. Find the critical points of f . / Bereken die kritieke punte van f .

(a) $f(x) = 8x^2 - \ln(2x)$ (b) $f(x) = \frac{x^2}{\sqrt{x+1}}$

5. Sketch / Skets $f(0) = 2, f(4) = 3, f'(x) < 0$ if/as $x > 2, f'(x) > 0$ if/as $x < 2,$

$$f''(x) < 0 \text{ if/as } x \in [-1, 4], f''(x) > 0 \text{ if/as } x < -1 \text{ or } x > 4$$